

Simulating Photoinduced Electron Dynamics for X-ray Absorption Spectroscopy and X-ray Natural Circular Dichroism in Molecules

Manuel Fernando Sánchez Alarcón^{1,2}, Emanuele Coccia³, and Majed Chergui^{2,4}

¹Dipartimento di Fisica, University of Trieste, Italy ²Elettra Sincrotrone, Trieste, Italy ³Dipartimento di Scienze Chimiche e Farmaceutiche, University of Trieste, Italy ⁴École Polytechnique Fédérale de Lausanne, Switzerland



Introduction

Chirality is central to biochemistry and drug design [1]. X-ray sources now extend chiroptical spectroscopy to core excitations, where resonant X-ray absorption (XAS) combines site specificity with strongly enhanced circular dichroism (XNCD) [2]. We address the need for theoretical assignments by propagating the time-dependent Schrödinger equation under an explicit pulse to compute XAS and XNCD in molecules with WaveT [3], enabling interpretation of X-ray chiroptical spectra and ground- and valence excited-state pump-probe simulations [4].

Methods

Real-time propagation (WaveT): XAS and XNCD are obtained from the time-dependent Schrödinger equation in the length gauge or the velocity gauge [5]:

$$i\frac{d}{dt}|\Psi(t)\rangle = (\hat{H}_0 + \hat{H}_I(t))|\Psi(t)\rangle, \quad (1)$$

where $\hat{H}_I(t)$ is the interaction Hamiltonian for the length gauge ($\hat{H}_I(t) = -\hat{\mu} \cdot \vec{F}(t)$) or the velocity gauge ($\hat{H}_I(t) = \hat{p} \cdot \vec{A}(t)$), and driven by a Gaussian-enveloped oscillatory pulse $\vec{F}(t) = \vec{F}_m \exp\left(-\frac{(t-t_0)^2}{2\sigma^2}\right) \sin(\omega_c t) = -\frac{\partial \vec{A}(t)}{\partial t}$, with carrier frequency ω_c , center t_0 , width σ , and peak amplitude $\vec{F}_m$.

State-space expansion: $|\Psi(t)\rangle$ is expanded in TDDFT eigenstates of $\hat{H}_0$. For ground-state (GS) spectra (core-valence separation, CVS):

$$|\Psi(t)\rangle = C_0(t)|0\rangle + \sum_{\lambda_c=1}^{N_c} C_{\lambda_c}(t)|\lambda_c\rangle. \quad (2)$$

For excited-state (ES) spectra:

$$|\Psi(t)\rangle = C_0(t)|0\rangle + \sum_{\lambda_c=1}^{N_c} C_{\lambda_c}(t)|\lambda_c\rangle + \sum_{\lambda_v=1}^{N_v} C_{\lambda_v}(t)|\lambda_v\rangle. \quad (3)$$

Spectra (linear response): With $\Delta\mu_q(t) = \mu_q(t) - \mu_q(0)$, $\Delta m_q(t) = m_q(t) - m_q(0)$ the time-dependent induced dipole moment and magnetic moment, along axis q , and $F_n^0(\omega)$ the Fourier transform of the field along axis n , the isotropic absorption and circular dichroism spectra are

$$P^A(\omega) = \frac{1}{3} \sum_{n \in \{x,y,z\}} P_{nn}^A(\omega), \quad P^{CD}(\omega) = \frac{1}{3} \sum_{n \in \{x,y,z\}} P_{nn}^{CD}(\omega),$$

$$P_{nq}^A(\omega) = \frac{1}{2\pi F_n^0(\omega)} \int_0^\infty \Delta\mu_q(t) e^{i(\omega+i\Gamma)t} dt, \quad P_{nq}^{CD}(\omega) = \frac{i}{2\pi\omega F_n^0(\omega)} \int_0^\infty \Delta m_q(t) e^{i(\omega+i\Gamma)t} dt, \quad (4)$$

where Γ is a damping parameter (average excited-state lifetime).

Simulation pipeline:

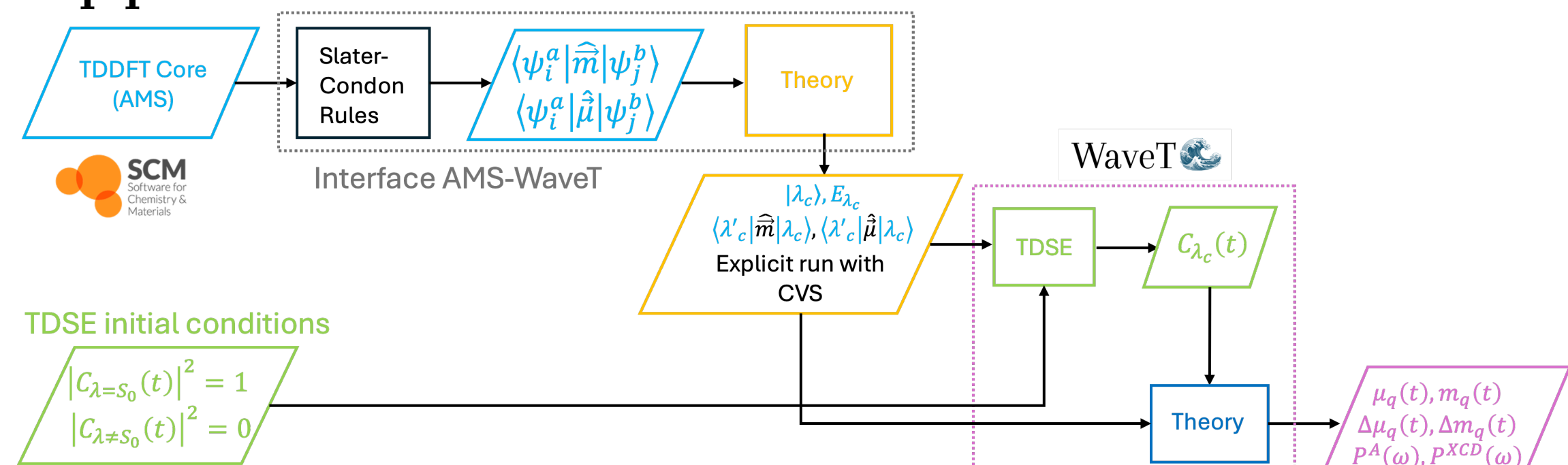


Figure 1: Pipeline for the computation of ground-state XAS and XNCD in molecules.

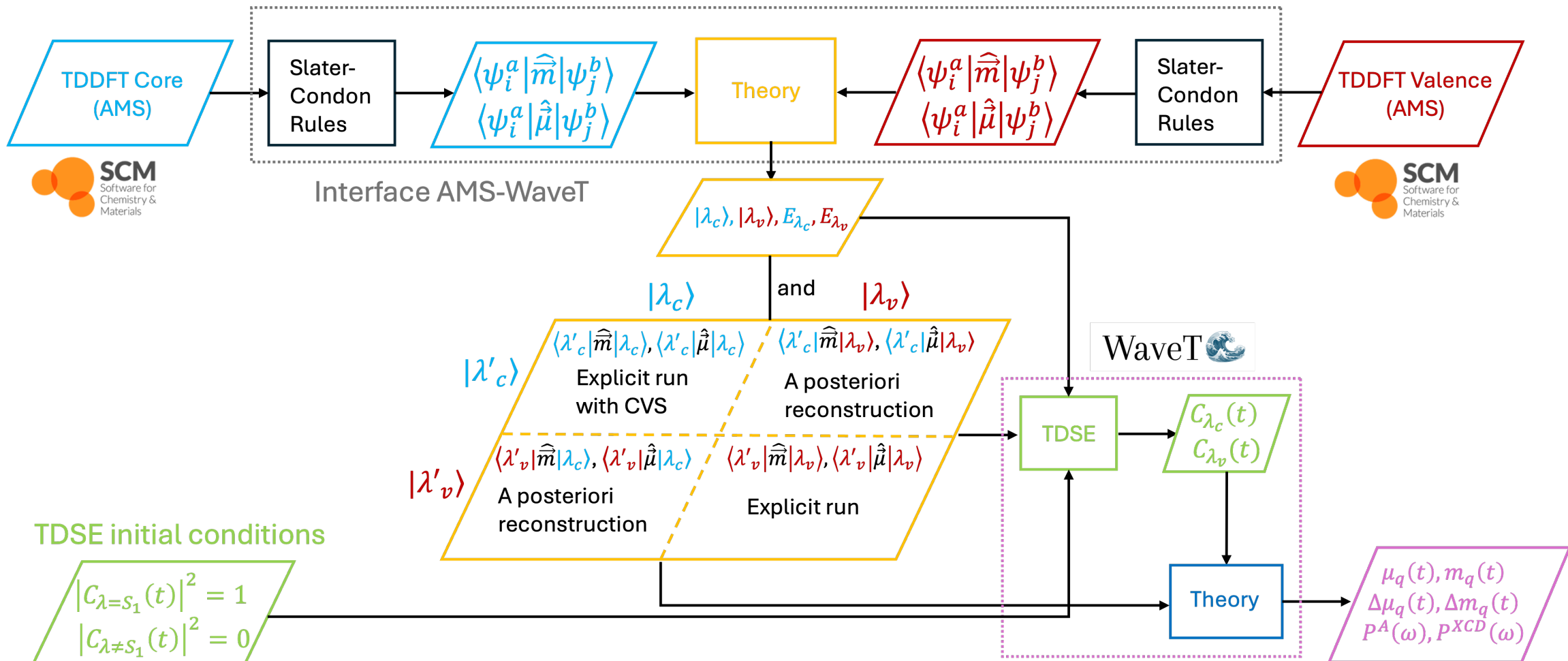


Figure 2: Pipeline for the computation of excited-state XAS and XNCD in molecules.

Computational Details

System and theory: (R)-methyloxirane; geometry from Andersen *et al.* [6] (reference for comparison). Ground state TDDFT simulations: DFT with ADF/AMS, CAM-B3LYP.

Basis sets (BS1–BS4): C K-edge and valence results were benchmarked to choose an efficient, converged setup: BS1: C,O augmented TZP; H TZP. BS2: C QZP+1 diffuse; O,H TZP. BS3: C,O QZP+1 diffuse; H DZP. BS4: C,O QZP+1 diffuse; H TZP.

Core and valence states: Core TDDFT under CVS (60 states retained). Valence TDDFT without orbital freezing (60 states retained). Real-time dynamics at the C K-edge: 17 core states and 60 valence states; core manifold limited to excitations below the C 1s ionization threshold.

Propagation: 97 fs trajectories with time step $\delta t = 0.12$ as, field intensity $I = 10^4$ W cm⁻², $\omega_c = 278$ eV (C K-edge XAS/XNCD), pulse FWHM 1 fs, $\tau = 1/\Gamma = 200$ a.u.

Results and Discussion

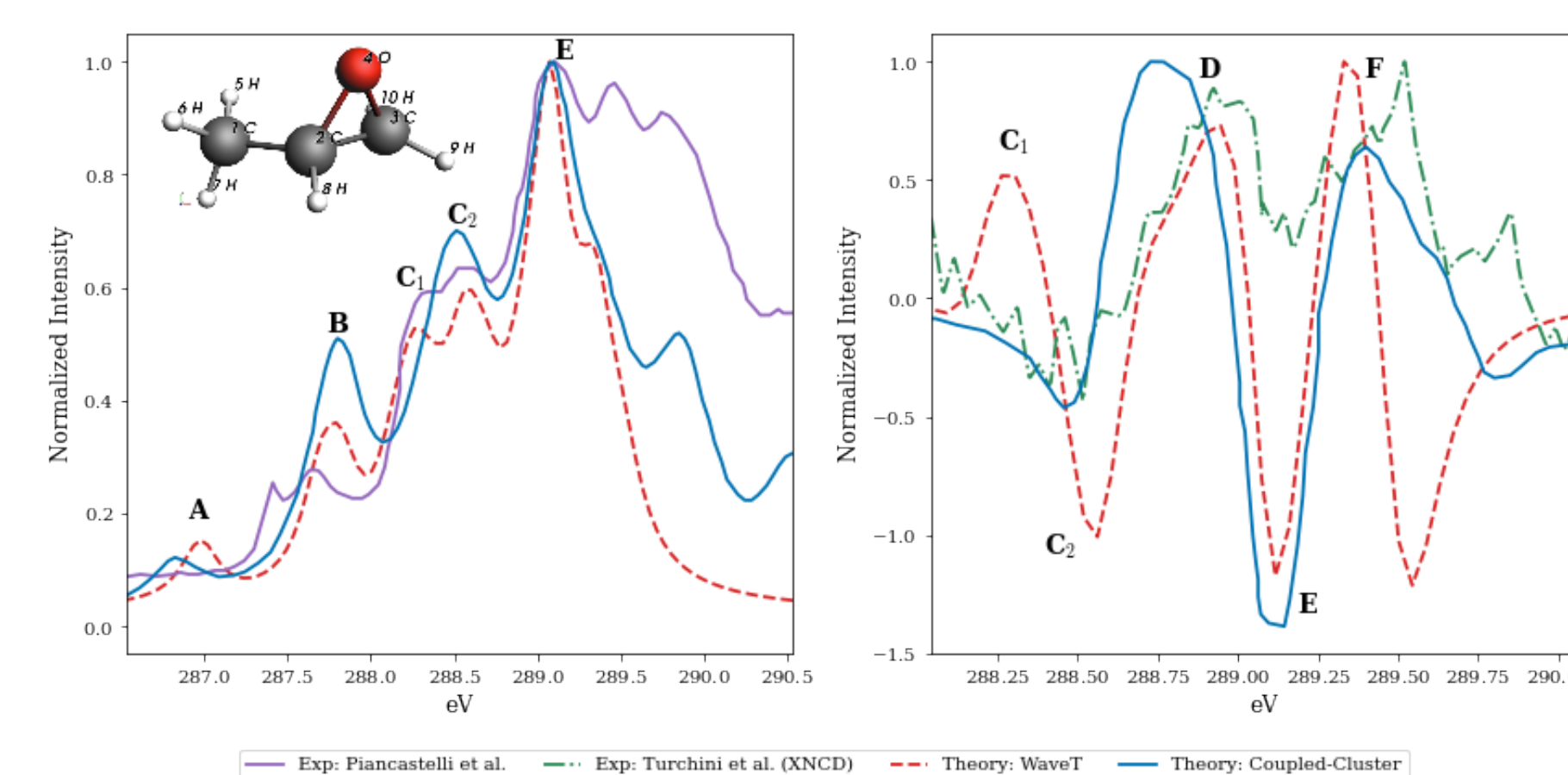


Figure 3: Experimental vs. computed ground-state XAS and XNCD for (R)-methyloxirane. Couple-Cluster data was obtained from Andersen *et al.* [6], while experimental data was obtained from [7, 8].

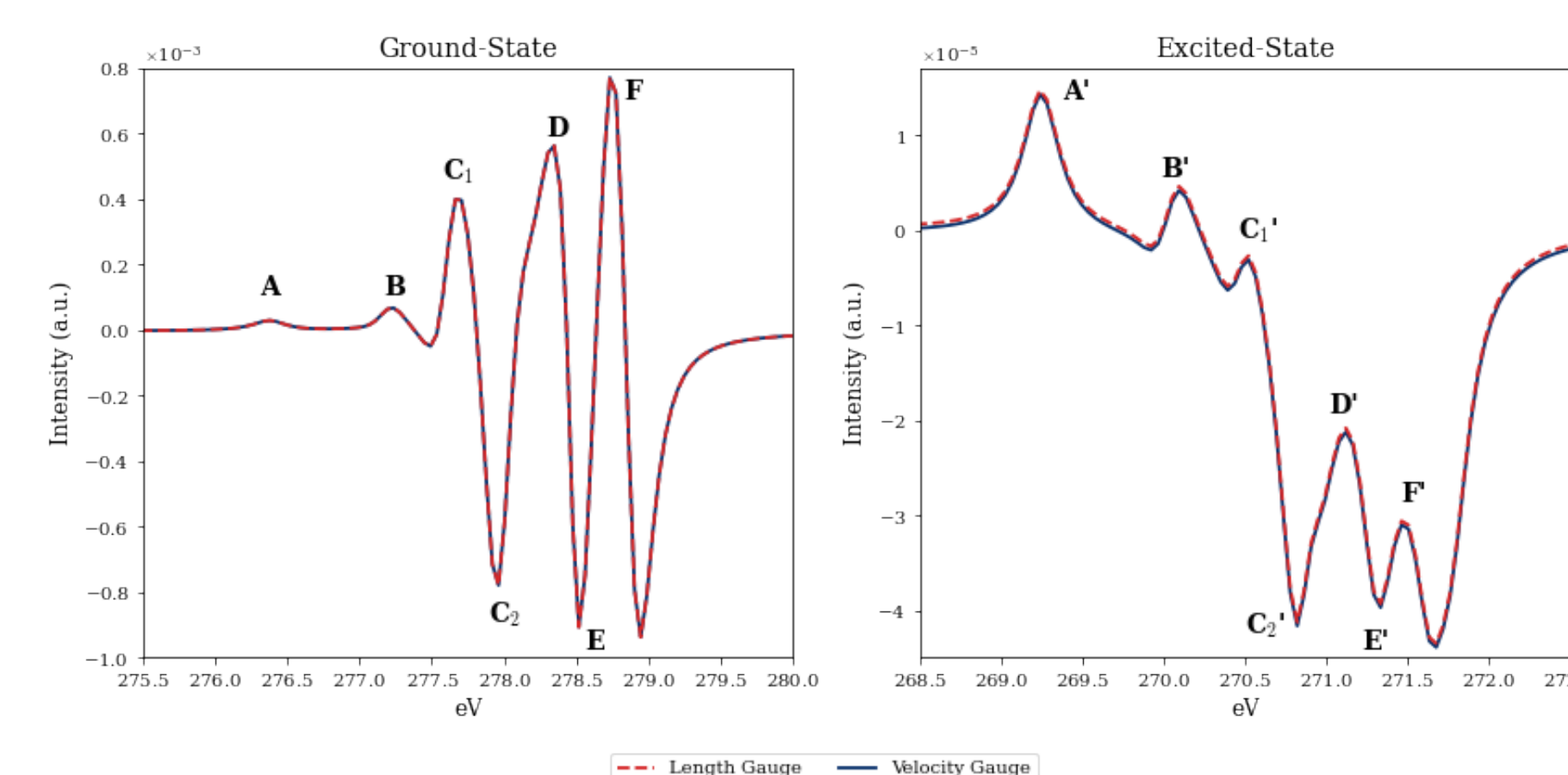


Figure 4: Computed first-excited-state XNCD for (R)-methyloxirane.

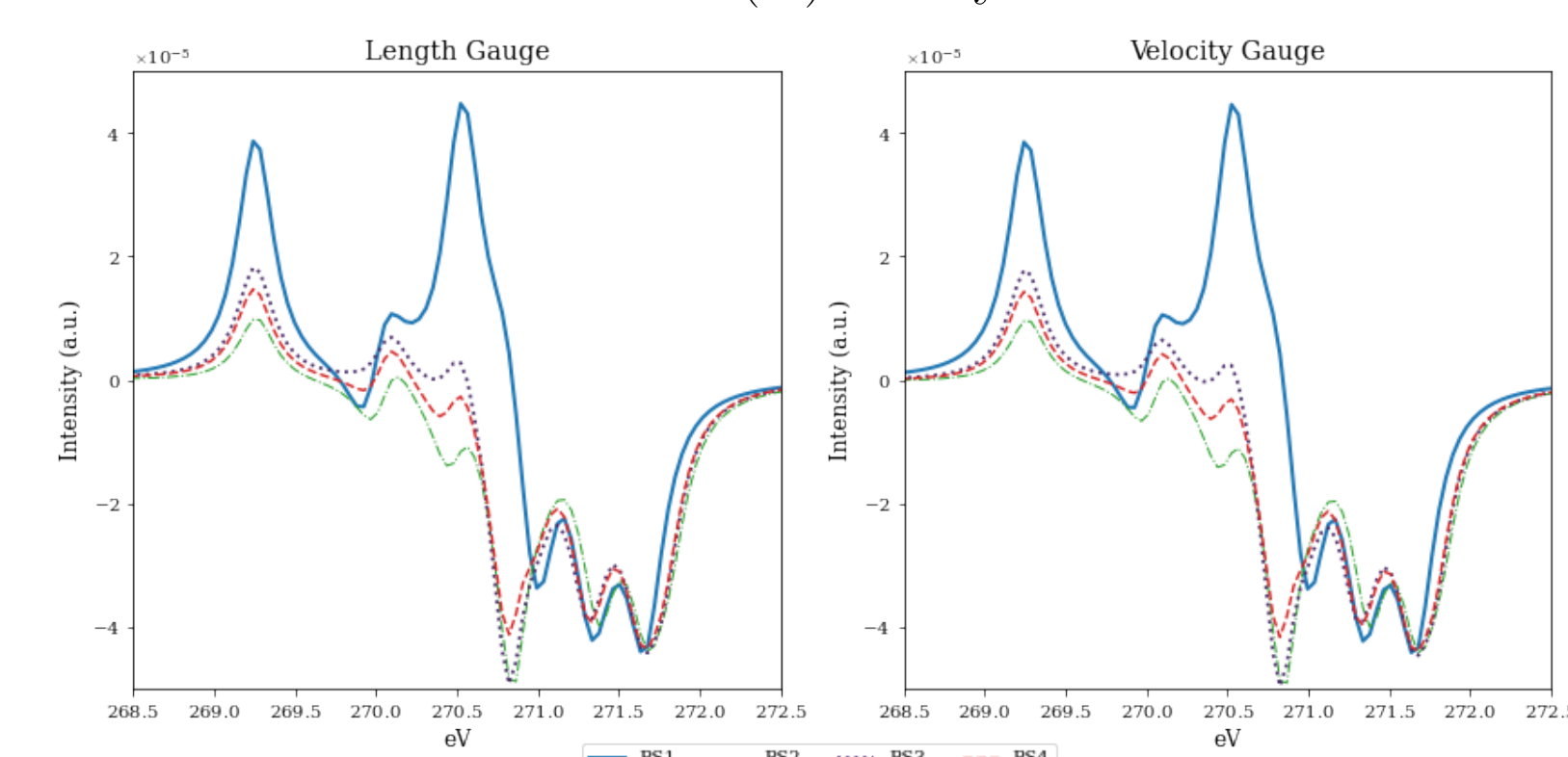


Figure 5: Basis set convergence of computed first-excited-state XNCD for carbon K-edge of (R)-methyloxirane. Core and valence calculations used to obtain the excited-state XNCD share the same basis set.

Conclusions

- WaveT is a general framework for molecular XAS and XNCD, including excited-state spectra and prospective solvent effects (perturbative matrix method or COSMO).
- For (R)-methyloxirane, WaveT ground-state XAS and XNCD reproduce experiment and agree with coupled-cluster benchmarks.
- Ground- and excited-state core transitions are assigned unambiguously.

References

- ¹M. Chergui, *Structural Dynamics* **3**, 031001 (2016).
- ²M. Tanaka *et al.*, *Physica Scripta* **2005**, 873 (2005).
- ³M. Monti, M. Stener, and E. Coccia, *The Journal of Chemical Physics* **158**, 084102 (2023).
- ⁴M. F. Sánchez, E. Coccia, and M. Chergui, *In preparation*.
- ⁵S. Pistillo *et al.*, *In revision*.
- ⁶J. H. Andersen, K. D. Nanda, A. I. Krylov, and S. Coriani, *Journal of Chemical Theory and Computation* **18**, 1748 (2022).
- ⁷M. Piancastelli *et al.*, *Journal of Electron Spectroscopy and Related Phenomena* **156-158**, 259 (2007).
- ⁸S. Turchini *et al.*, *Journal of the American Chemical Society* **126**, 4532 (2004).

Acknowledgements

We thank ... for financial support.